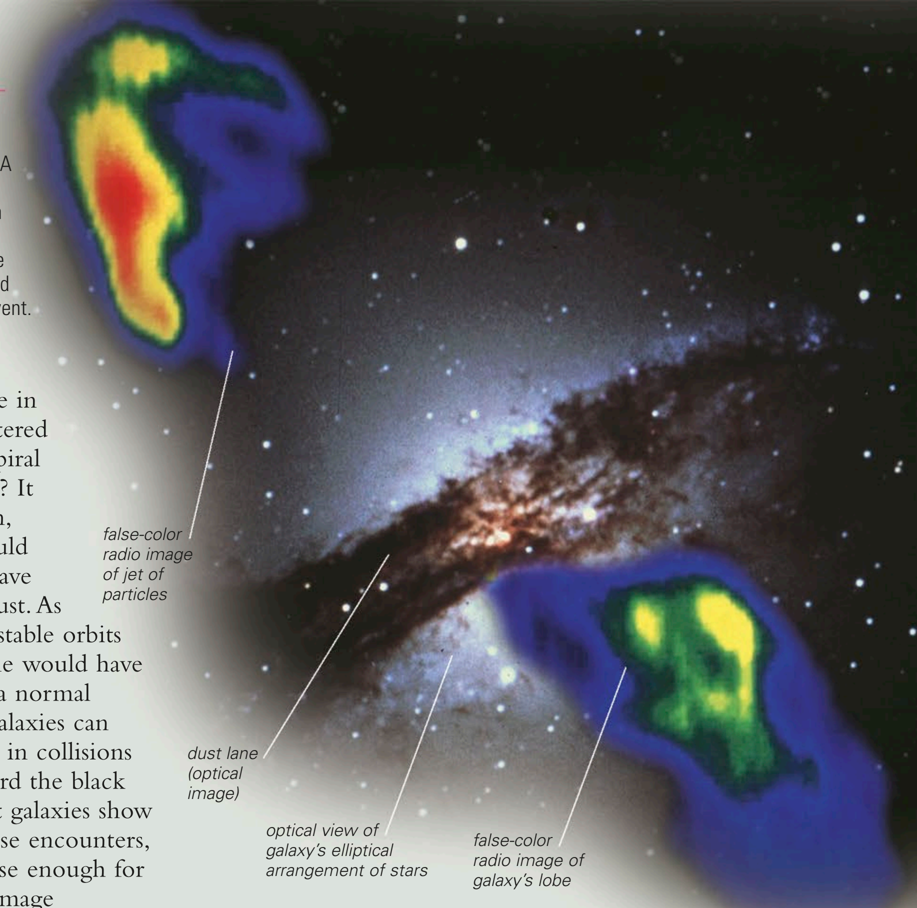
THE HISTORY OF ACTIVE GALAXIES

The distribution of different types of active galaxies in the universe provides clues about how they evolve. Quasars and blazars are never seen close to Earth. They are always faint and distant, with red shifts indicating that they lie billions of light-years from Earth—we are seeing them as they were in much earlier times. Radio and Seyfert galaxies, in contrast, are scattered throughout the nearby universe, and radio jets are linked to both spiral and elliptical galaxies. So what happened to the quasars and blazars? It seems likely that they represent a brief phase in a galaxy's evolution, soon after its birth. At this time, material in the central regions would have had chaotic orbits, and the central black hole engine would have been fueled by a continuous supply of infalling stars, gas, and dust. As the black hole swept up the available matter, objects with stable orbits at a safe distance remained. Starved of fuel, the engine would have petered out, and the quasar became dormant—a normal galaxy such as the Milky Way. Today, such galaxies can become active again if they are involved in collisions that send new material falling in toward the black hole. Many nearby radio and Seyfert galaxies show evidence of recent collisions or close encounters, and some of these galaxies are close enough for infrared telescopes to image the dust rings around their cores directly (see p.323). However, levels of recent activity are restrained—even the most spectacular radio galaxies generate little energy compared to quasars, while Seyferts are the feeblest type of active galaxy.



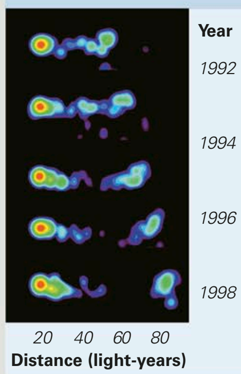
NUDGED BACK INTO LIFE
Optical images of Centaurus A clearly show the dark dust lane of a spiral colliding with this elliptical galaxy. The overlaid radio map shows the burst of activity—the jets and plumes—triggered by this event.

ACTIVE GALAXY
This idealized active galaxy is a spiral with a bright nucleus that hides an active black hole. From the black hole's poles blast two jets of particles, leaving at close to light speed, only slowing and billowing out into lobes many thousands of light-years away as the particles hit the intergalactic medium.

EXPLORING SPACE

SUPERLUMINAL JETS

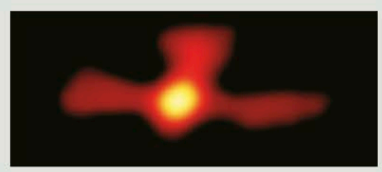
Some quasars and blazars appear to defy the laws of physics. Image sequences, taken years apart, show jets of material blasting away from the nucleus, apparently traveling faster than the speed of light. This apparent motion is called "superluminal." In reality, it is an illusion created when jets traveling at very high speeds, of up to 99 percent of the speed of light, happen to be pointing almost directly toward us.



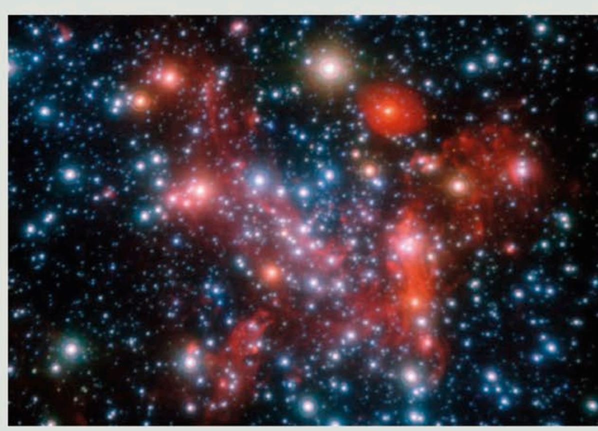
TIME-LAPSE SEQUENCE
These images show jet emissions from blazar 3C 279, taken at intervals of almost 2 years, and showing motion apparently five times the speed of light.

IS THE MILKY WAY ACTIVE?

The Milky Way Galaxy, like any galaxy with a central black hole, has the potential to be active, and there is intriguing evidence that it might have burst into activity in the recent past. In 1997, scientists discovered a huge cloud of gamma-ray emission above the galactic center. The radiation has a distinctive frequency, suggesting it is the result of electrons encountering positrons—their antimatter equivalent (see p.31)—and annihilating in a burst of energy. The positrons might have been generated by activity at the core—perhaps an infall of matter into the black hole—and are now meeting scattered electrons in the outer galaxy and mutually annihilating to produce the distinctive glow. Because the clouds lie just 3,000 light-years from the galactic center, the activity must have occurred recently.



ANTIMATTER FOUNTAIN
This gamma-ray image traces positrons (antielectrons) around the Milky Way. The horizontal feature is the plane of the galaxy, with the fountain above it.



GALACTIC CENTER
This near-infrared image, taken using the Very Large Telescope in Chile, shows the center of the Milky Way. By following the motions of its central stars over more than 16 years, astronomers were able to determine that the supermassive black hole at the core is about 4 million times as massive as the Sun.